

MERN Stack Web Development Internship Report

Name: Priya Ashokbhai Rana

Duration: 14 May, 2026 to 14 June, 2026

Organization: Birla Vishvakarma Mahavidyalaya

Technologies: Frontend- React.js, Tailwind CSS

Backend- Node.js, Express.js

Database- MongoDB

Tools- GitHub, Postman, VS Code

Organization: TechnoGuide InfoSoft Pvt. Ltd.

Acknowledgement: I want to thank my mentors for giving me this chance to complete the 21-day MERN Stack internship. This internship helped me improve my full stack development skills and gave me practical experience in web development.

Objective of Internship:

- Learn full stack web development using MERN stack
- Understand frontend and backend integration
- Build real-world projects
- Improve problem-solving and coding skills
- Learn GitHub and deployment process
- Learn about rest APIs.

Daily Work Summary:

Week 1: Frontend Fundamentals

- Day 1: Environment setup, GitHub repo creation
- Day 2: HTML & CSS responsive registration page
- Day 3: JavaScript basics + Calculator app
- Day 4: DOM, Local Storage + To-Do App
- Day 5: Bootstrap/Tailwind portfolio page
- Day 6: React introduction + Profile Card
- Day 7: React state + Counter & Toggle app

Week 2: React & API Integration

- Day 8: Forms & validation (Login/Signup UI)

- Day 9: React Router multi-page website
- Day 10: API integration (fetch users)
- Day 11: CRUD React project (Student Management UI)
- Day 12: Node.js basics
- Day 13: Express.js REST API
- Day 14: MongoDB CRUD operations

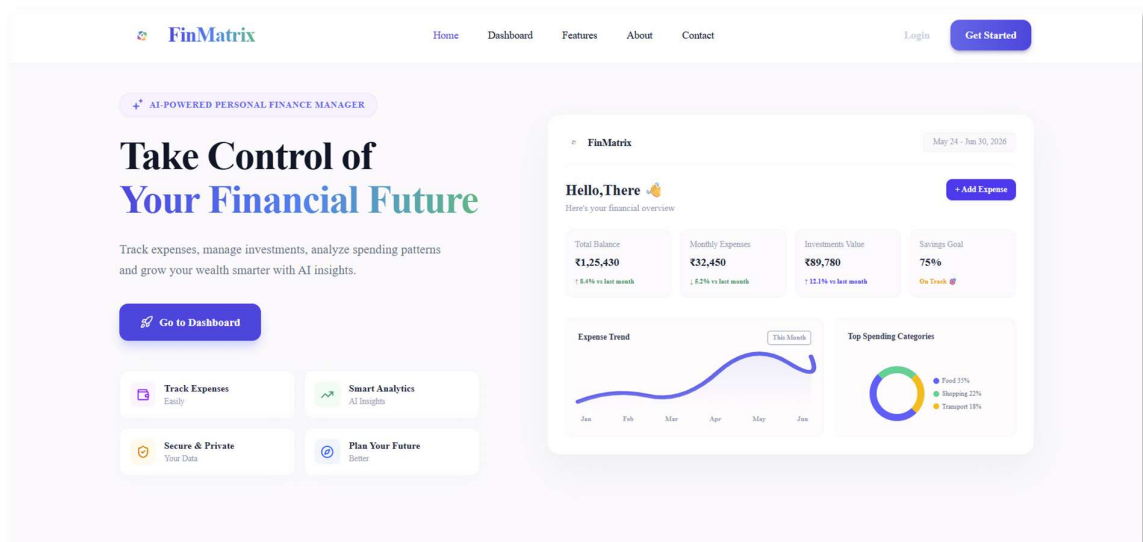
Week 3: Backend & Full Stack Project

- Day 15: Mongoose integration
- Day 16: Backend CRUD APIs
- Day 17: JWT Authentication system
- Day 18: MERN full stack integration
- Day 19: File upload & deployment basics
- Day 20: Final project development
- Day 21: Deployment & final presentation

Final Project:

FinMatrix:

FinMatrix is a modern financial dashboard web application that helps users manage their personal finances in an organized way. It provides an easy-to-use interface for tracking transactions, managing bank accounts, setting financial goals, and getting AI-based insights for better decision making.



Key Features:

1. Overview Dashboard

- Real-time summary of income, expenses, and savings
- Day / Month / Year filtering for analytics

2. Transactions Module

- Add, view, and manage all financial transactions
- Categorized expense tracking

3. Bank Accounts Section

- Manage multiple bank accounts in one place
- Track account balances

4. Goal Management

- Set financial goals (saving targets)
- Track progress visually

5. AI Insights (Important for resume)

- Smart suggestions based on spending pattern

6. Clean Analytics UI

- Graph-based financial visualization
- Easy understanding of money flow

7. User-Friendly Dashboard

- Simple navigation sidebar (Overview, Transactions, etc.)
- Responsive and modern UI design

Learning Outcomes:

- Learned full MERN stack development
- Understood frontend-backend communication
- Built and deployed full stack applications
- Improved GitHub version control skills
- Gained real-world project experience

Conclusion:

This MERN Stack internship helped me understand practical web development and build real projects. It improved my technical skills and boosted my confidence for future internships and job opportunities.